

Supporting Information

For

Graph Symmetry and Kekulé Structure Localization Correlate with Carcinogenic Potency of Polycyclic Aromatic Hydrocarbons

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Contents

- Table S1. PubChem confirmations and graph-verification notes for corrected or high-risk structures (extended commentary on five cases requiring explicit annotation).
- Table S2. Complete 27-PAH descriptor table used in the main analysis. The CID column lists the canonical PubChem identifier for every entry; all 27 structures were programmatically retrieved from PubChem and the resulting molecular graphs were independently verified before being passed to the K and |Aut(G)| computations.
- Table S3. Sensitivity analyses for floor assignment and benzo[c]phenanthrene proxy assignment.
- Table S4. Leave-one-out predictions and residuals for every molecule. The three methylated PAHs (3-methylcholanthrene, 5-methylchrysene, and 7,12-dimethylbenz[a]anthracene) — the systematic failure mode discussed in the main text — appear with LOO residuals of +1.30, +1.78, and +2.41 log₁₀ units respectively, all in the direction of underprediction; this is the basis for the decomposition RMSE_{unsubstituted} = 1.15 vs RMSE_{methylated} = 1.88.
- Table S5. Dibenzopyrene isomer comparison, including the qualitative comparator dibenzo[e,l]pyrene.
- Figure S1. ROC curve for the binary classification task at threshold K/|Aut(G)| ≥ 5.0 across the full 27-PAH dataset (full-set AUC = 0.95; EPA Priority subset n=14 AUC = 0.95). Exported as figS1_ROC.pdf and figS1_ROC.png.

Selection of the binary threshold $K/|Aut(G)| \geq 5.0$

The threshold was selected at the natural gap in the descriptor distribution (Table S2): dibenz[a,h]anthracene ($K/|Aut| = 5.00$, PEF = 5.0) is the lowest- $K/|Aut|$ compound classified as a regulatory carcinogen in the primary hierarchy and serves as the anchor point. Shifting the cutoff by ± 0.5 around 5.0 trades a single boundary case in each direction without improving the overall confusion matrix; the value 5.0 was therefore retained as the simplest defensible cut. The two false positives (benzo[ghi]perylene, benzo[e]pyrene) and two false negatives (benzo[k]fluoranthene, dibenz[a,h]anthracene) are explicitly retained and discussed in the main text rather than tuned away.

Key SI Notes

- All 27 structures are cross-verified against PubChem entries (Table S2, CID column). The five cases below required explicit annotation because they involved

nomenclature ambiguity, prior literature errors, or boundary failure modes; the remaining 22 structures matched their canonical PubChem entries on the first retrieval.

- 5-Methylchrysene is confirmed against PubChem CID 19427.
- 7,12-Dimethylbenz[a]anthracene (DMBA) is confirmed against PubChem CID 6001.
- Benzo[k]fluoranthene remains a real boundary false negative after pynauty confirmation of $|Aut| = 2$.
- Dibenzo[a,e]pyrene is corrected to PubChem CID 9126 with $K = 17$ and $|Aut| = 1$.
- Dibenzo[e,l]pyrene maps to PubChem CID 9122 and gives $K = 20$, $|Aut| = 4$, $K/|Aut| = 5.0$.

Exported Files

- Table S1
- Table S2
- Table S3
- Table S4
- Table S5
- Figure S1 PDF
- Figure S1 PNG

LOO Summary

- LOO Spearman $\rho = +0.708$ (95% bootstrap CI: +0.480 to +0.834)
- LOO RMSE = 1.25 log₁₀ units
- RMSE unsubstituted (n=24) = 1.15
- RMSE methylated (n=3) = 1.88
- LOO classification accuracy = 19/27 (70.4%)

Binary Summary

- Threshold $K/|Aut| \geq 5.0$
- TP = 11, TN = 11, FP = 2, FN = 2
- AUC = 0.95 (full dataset); AUC = 0.95 (EPA Priority PAHs only, n=14)

Bootstrap 95% CI

- Primary $\rho = +0.745$ (95% CI: +0.546 to +0.855, n=10000 resamples, seed=42)
- EPA-only (n=14): $\rho = +0.759$, $p = 0.0016$, AUC = 0.95